

Multiple Choice (10 marks)

1. In Python 3, what is `['a', 'Chemistry', 0, 1][1]`?
 - (a) a
 - (b) Chemistry
 - (c) 0
 - (d) 1
2. Which is the smallest asymptotically?
 - (a) $O(1)$
 - (b) $O(n)$
 - (c) $O(n^2)$
 - (d) $O(\log n)$
3. Let A be a finite set with m elements, and let B be a finite set with n elements. The number of distinct functions mapping A into B is
 - (a) n^m
 - (b) $n!/(n - m)!$
 - (c) $n!$
 - (d) $n!/(m!(n - m)!)$
4. In Python 3, what is the output of `['Hi!'] * 4`?
`'Hi!', 'Hi!', 'Hi!', 'Hi!'`
`'Hi!' * 4`
 - (a) Error
 - (b) None of the above.
5. Let $f(X) = \text{if } x = 1 \text{ then } 0 \text{ else } [x * f(x - 1) + x**2]$. The value of $f(4)$ is
 - (a) 53
 - (b) 29
 - (c) 50
 - (d) 100

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][-3]` evaluates to `'Chemistry'`.
7. True or False: All bit strings with more ones than zeros cannot be described with a regular expression.
8. True or False: Finding a longest simple cycle in an undirected graph G is known to be solvable in polynomial time.
9. True or False: The function `max(l)` returns the highest value in the list `l`, which is 4 for the list `[1, 2, 3, 4]`.
10. True or False: Merge sort has a running time that is least affected by the initial ordering of the input data compared to other sorting algorithms.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to extract all the adjacent coordinates of the given coordinate tuple.
12. Write a python function to check whether the given array is monotonic or not.

End of Paper